STAT 443 Notes

Thomas Liu

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1 Introduction to Forecasting, Control and Time Series

1.1 Time Series

1.1.1 Time Series Plot

- A time series plot is a scatterplot of the data with time on the x-axis in, typically, equally spaced intervals, and observations on the y-axis
- For visual clarity adjacent observations are connected by lines
- The equally spaced intervals can be in units of years, months, days, hours etc and these unites are called the **period** of the time series

1.1.2 Level, Trend and Seasonality

- The **level** is the local mean of the observations, around which we see random **noise**
- When the level varies with time we say there is a **trend**
- A **seasonal effect** is a systematic and calendar related eeffect which repeats with a given period

1.1.3 Change Point

A change point is a time at which at least of two of the following changes: the data generation process, the way that the data is measured, or the way that the observation is defined
Make the forecast very difficult, as we don't know what's going to happen after the change point

1.1.4 Stationarity / Non-stationarity

Seasonality, trends, non-constant variance and change points are all examples of non-stationarity
Stationarity means that the underlying random process does not change in time

1.2 Simple Statistical Tools

1.2.1 Model with Trend

A trend model for the time series X_t is a decomposition

$$X_t = m(t) + Y_t$$

where $m(t)$ is a function of time t and Y_t has zero mean. Commone examples are

$$m(t) = \alpha_0 + \alpha_1 t$$

and

$$m(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2$$

where α_i are unknown parameters
We want

1. $E(Y_t) = 0$ for all t
2. $Cov(Y_t, Y_s) = 0$ for all $t \neq s$, Y_t are independent
3. $Var(Y_t) = \sigma^2$ for all t, Y_t are identically distributed, variance constant
4. $Y_t \sim N(0, \sigma^2)$, Y_t are normally distributed

1.2.2 Model with Seasonal Component

A model with seasonal component with period d for X_t is a decomposition

$$X_t = s(t) + Y_t$$

where Y_t has zero mean but now $s(t)$ is periodic so statisfies $s(t) = s(t + d)$ for all t

1.2.3 Linear Decomposition Model

A linear decomposition model for the time series X_t is a decomposition

$$X_t = m(t) + s(t) + Y_t$$

where $E(Y_t) = 0$, $s(t)$ is periodic with period d and for identification reasons we further assume

$$\sum_{t=1}^{d} s(t) = 0$$

We can construct multiplicative decomposition model by applying a linear decomposition to $\log(X_t)$

1.2.4 Identification of a Model

A model is not identified if there is more than one way to write the same model. If there is a unique way we say the model is identified

Example

We have two periodic functions $s_1(t)$ and $s_2(t)$ with period d such that $s_1(t) = s_2(t) + 1$

$$X_t = s_1(t) + Y_t = s_2(t) + 1 + Y_t = s_2(t) + Y_t^*$$

where Y_t has zero mean but Y_t^* has mean 1

Decomposition Using R

We can use the following code to decompose a time series X_t in R

```
plot(decompose(birth, type = "additive"))
```

There are two types of decomposition, additive and multiplicative.
Within the output

- The top plot (observed) is the original time series X_t
- The second plot (trend) is the trend component $m(t)$
- The third plot (seasonal) is the seasonal component $s(t)$
- The fourth plot (random) is the residual component Y_t

We can have

$$X_t = m(t) + s(t) + Y_t$$

1.2.5 Simple Moving Average

If the period is odd $d = 2q + 1$ the the smoother is defined on the realisation x_t by

$$\hat{m}(t) = \frac{x_{t-q} + x_{t-q+1} + \cdots + x_{t+q}}{d}$$

and if even $d = 2q$ then

$$\hat{m}(t) = \frac{0.5x_{t-q} + x_{t-q+1} + x_{t-q+2} + \cdots + x_{t+q} + 0.5x_{t+q}}{d}$$

with 0.5 weights at the start and end to give a total weight of $d = 2q$

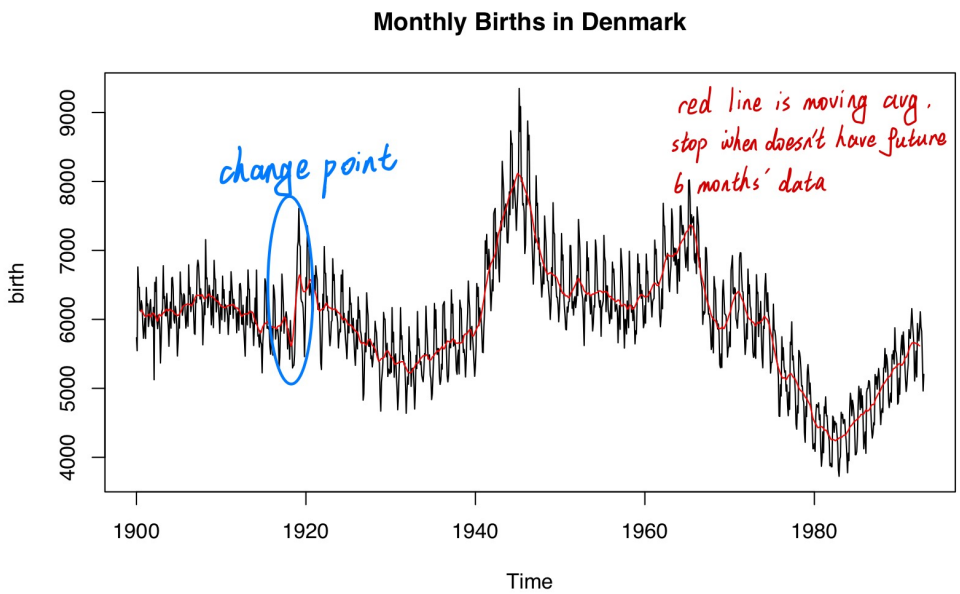
1.2.6 Simple Moving Average Filter

We can estimate the seasonal components. For each $k = 1, \cdots, d$ compute the average w_k of

$$x_{k+jd} - \hat{m}_{k+jd} | q < k + jd \leq n - q$$

We then normalise to get

$$\hat{s}_k = w_k - \frac{\sum_1^d w_j}{d}$$



1.2.7 Exponential Smoothing

- Exponential smoothing can be used for a time series to estimate the level of the process m_t which is assumed slowing varying
- It should not be used when there is a trend or seasonality
- Assume that we can observe $x_1, \cdots, x_n$. We select an initial value $\hat{m}(0)$, typically x_1 then we update the estimate recursively via the update equation

$$\hat{m}(t+1) = \alpha x_{t+1} + (1-\alpha)\hat{m}(t)$$

  for $\alpha \in (0,1)$ which mixes the current estimate of $\hat{m}(t)$ with the value of x_{t+1} when it is observed

- The same update can be written in error correcting version as

$$\hat{m}(t+1) = \hat{m}(t) + \alpha(x_{t+1} - \hat{m}(t))$$

  is called the one-step ahead forecast error. The tuning parameter α needs to be selected and this can be done by finding the α which minimises

$$\sum_{t=1}^{T-1} (x_{t+1} - \hat{m}_\alpha(t))^2$$

  over the observed time period $\{1, \cdots, T\}$ and $\hat{m}_\alpha$ is the smoother for a fixed value of α

1.2.8 Holt-Winters Filtering

- The Holt-Winters method genralises exponential smoothing to the case where there is a trend and seasonality
- We have three terms which depend on time t
- The first a level a, the second is the trend b and the third is the seasonal component s with period p. The mean of the time series at time t is given by

$$m(t) = a + bt + s(t)$$

- The Hole-Winters prediction function for h time periods ahead of current time t is

$$\hat{x}(t+h) = \hat{a}(t) + \hat{b}(t)h + \hat{s}(t+h)$$

- The one step error term for this prediction is then defined as

$$\hat{e}(t) = x_t - (\hat{a}(t-1) + \hat{b}(t-1) + \hat{s}(t-p))$$

  then starting with initial values for a, b and s and then update the parameters according to the size of the error over the range of t by using

$$\hat{a}(t) = \hat{a}(t-1) + \hat{b}(t-1) + \alpha\hat{e}(t)$$

$$\hat{b}(t) = \hat{b}(t-1) + \alpha\beta\hat{e}(t)$$

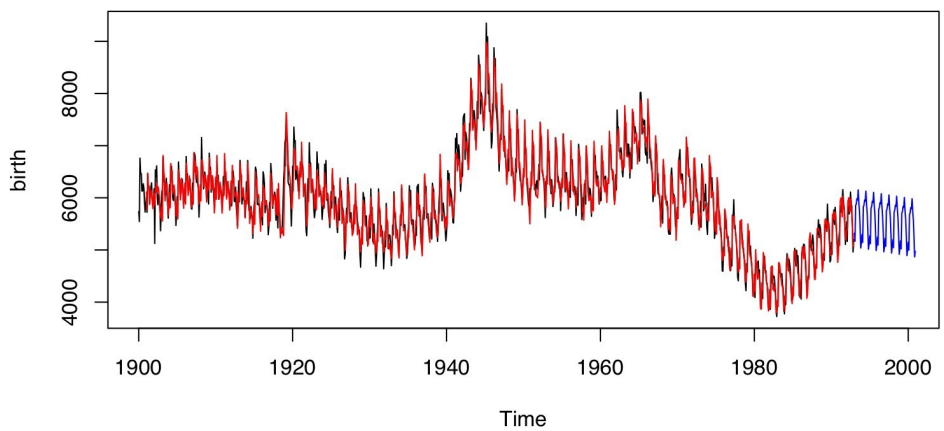
$$\hat{s}(t) = \hat{s}_{t-p} + \gamma\hat{e}(t)$$

- The method uses three tuning parameters α, β and γ and these are selected by minimising the sume of squared errors as in the case of exponential smoothing

We can use the following code to implement Holt-Winters in R

```
birth.hw <- HoltWinters(birth)
birth.hw.predict <- predict(birth.hw, n.ahead=12*8)
```

Which predict the next 8 years of monthly births in Denmark



- The black line is the original time series X_t
- The red line is the fitted Holt-Winters model $m(t)$
- The blue line is the forcasting for the next 8 years, $h = 12 * 8$ months

2 Regression Methods and Model Buiding Principles

2.1 Linear Regression

We can model with time series with a trend using model

$$Y_t = \alpha_0 + \alpha_1 t + \epsilon_t$$

where $E(\epsilon_t) = 0$
Using Ordinary Least Squares estimation, we can estimate $\hat{\alpha}_0$ and $\hat{\alpha}_1$ by minimising

$$\sum_{t=1}^{n} (Y_t - \hat{\alpha}_0 - \hat{\alpha}_1 t)^2$$

We can also estimate the standard deviation

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^{n} (Y_t - \hat{\alpha}_0 - \hat{\alpha}_1 t)^2$$

We can predict a value Y_{t^*} for some time t^* using the model

$$\hat{Y}_{t^*} = \hat{\alpha}_0 + \hat{\alpha}_1 t^*$$

The uncertainty measeaured by the standard error is

$$\sqrt{\hat{\sigma}^2 (1 + \frac{1}{n} + \frac{(t^* - t)^2}{\sum_t (t - \bar{t})^2})}$$

2.1.1 Important Lessons

- Making a prediction where the new covariate values are far from the values used to fit the data is very risky
- Goodness of fit only tells the model fits the training data and not much/very little about how model behaves when far from the training data
- We also used data multiple times during the procedure (model selection, model fitting, prediction, and assessment of prediction uncertainty). This can give rise to both bias in the forecast and poor estimation of the prediction uncertainty
- We also need to make sure that the errors are not correlated

2.2 Prediction in Multivariate Regression

We make the following assumptions

$$E(Y|X) = \beta_0 + \sum_{i=1}^{p} \beta_i X_i$$

$$Y_i = \beta_0 + \sum_{i=1}^{p} \beta_i X_i + \epsilon_i$$

Assume

$$\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$$

Gauss-Markov conditions
We use the matrix notation

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

We write

$$Y = X\beta + \epsilon$$

To find the estimate for β, we want to minimise the residual sum of squares

$$RSS(\beta) = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{N} (y_i - X_i\beta)^2 = \sum_{i=1}^{N} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2$$

2.2.1 OLS Estimation

The ordinary least squares estimator is given by

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

when $(X^T X)^{-1}$ exists

2.2.2 Variance of OLS

The variance-covariance of the sampling distribution of $\hat{\beta}$ is given by

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

2.2.3 Multicollinearity

If two, or more, of the columns of X are highly correlated, we say we have multicollinearity

2.2.4 Prediction Interval in Regression

If we assume that (Y_0, X_0) is independent of the training set $T = \{(Y_1, X_1), \cdots, (Y_n, X_n)\}$, and assume we have a correctly specified model
Define the point forecast as

$$\hat{\mu} = x_0^T \hat{\beta}$$

Then a $\alpha\%$ prediction interval is

$$\hat{\mu} \pm c_\alpha \hat{\sigma} \sqrt{1 + x_0^T (X^T X)^{-1} x_0}$$

where c_α is the $1 - \frac{\alpha}{2}$ quantile from the t_{N-p-1} distribution
The size of the term

$$x_0^T (X^T X)^{-1} x_0$$

can dominate the uncertainty in the forecast
If $X_T X$ is close to being singular - which is high multicollinearity - then, for certain values of x_0 this can be extermely large
If $X_T X$ is close to being singular, meaning small eigenvalue, which means the inverse will be large, causing the prediction interval to be large

2.3 Decision Theory

Let $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}$ be vector valued random variables where we are trying to predict Y given the information available in X

2.3.1 Loss Function

A loss function $L(Y, f(X))$ is a real-valued function which measures the error in estimating Y with the estimate $f(X)$

- Typically we assume $L(Y, f(X)) \geq 0$ with equality if and only if $y = f(x)$
- Since $L(Y, f(X))$ is a random variable, we may also consider its expected value, $E_{X,Y}[L(Y, f(X))]$, often called the risk
- The expectation is over the joint distribution of X, Y

A commonly used example is the squared error loss, or quadratic loss

$$L_{SE}(Y, f(X)) = (Y - f(X))^2$$

The expected value is called the mean square error (MSE)

$$MSE(f) = E_{X,Y}(L_{SE}(Y, f(X))) = E_{X,Y}((Y - f(X))^2)$$

Theorem

If X and Y are two random variables, with $E(Y) = \mu$ and $Var(Y) < \infty$. The function, f, which minimises $MSE(f)$, for X, Y, is given by the conditional expectation

$$f(X) = E_{Y|X}[Y|X]$$

- Mean square error is not the only possible loss function (easiest mathematically)
- We can alternatively use the absolute loss

  $$L_{abs}(Y, f(X)) = |Y - \hat{Y}(X)|$$

  where the 'best' esetimate is the conditional median rather than mean

2.4 Model Complexity

2.4.1 Bias-Variance Decomposition

Let T be the training set used to estimate $\hat{\beta} = \hat{\beta}(T)$, we now want to estimate the forecasting error for a new set of covariates, x_0. Let the forecast be

$$\hat{y}_0 = x_0^T \hat{\beta}$$

If y_0 is the actual value we are trying to forecast, and using quadratic loss we evaluate the forecast using

$$MSE(x_0) = Bias^2(\hat{y}_0) + Var_T(\hat{y}_0)$$

2.4.2 Expected Prediction Error

Suppose we have a regression model of the form

$$y = X\beta + \epsilon \quad Var(\epsilon) = \sigma^2$$

and a training set T from which we have estimated $\hat{\beta}(T)$. We want to assess the performace of the regression model for a new case with explanatary variates x_0 in terms of the bias-variance decomposition
Define expected prediction error as

$$EPE(x_0) = \sigma^2 + Bias^2 + Variance$$

The first term is called the irreducible error associated with the true model, f and will always be there, the second and third terms depend on the model class we use

2.4.3 Best Subset Regression

Best subset regression finds for wach $k \in \{0, 1, \cdots, p\}$, the subset which has the smallest residual sum of squares

2.4.4 Forward Step Selection

- This is a greedy algorithm which starts with the intercepts then adds the covariate which most decreases the RSS at each stage.
- Rather than explore the whole of the space of possible models it looks for a 'good' yet easy to compute path through them

2.4.5 Backward Step Selection

- Starts with the full model and deletes the covariate with the smallest z-statistic

2.4.6 Akaike Information Criterion (AIC)

- The AIC tries to balance goodness of fit of a model and the models complexity
- Measures the complexity of the model by counting the number of parameters used
- Defined as

$$-2l(\hat{\theta}) + 2N_p$$

  where l is log-likelihood, $\hat{\theta}$ the MLE and N_p the number of parameters in model
- Usually a smaller value of AIC is preferred

2.4.7 Lasson

- The lasso regression estimate is defined by

$$\hat{\beta}^{lasso} := \arg\min_{\beta} \sum_{i=1}^{N} (y_i - \beta_0 + \sum_{j=1}^{p} x_{ij}\beta_j)^2 \quad s.t. \sum_{j=1}^{p} |\beta_j| \leq t$$

- The constraint is on the absolute value of the parameter values not the squares. This can then also be expressed, using Lagrange multipliers

$$\hat{\beta}^{lasso} := \arg\min_{\beta} \left\{ \sum_{i=1}^{N} (y_i - \beta_0 + \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

3 Stationary Processes

3.1 Introduction

3.1.1 Time Series and Finite Dimensional Distributions

- A discrete time series is a set of random variables, $\{X_t\}$, indexed by $t \in T \subseteq \mathbb{Z}$
- For any finite subset of $\{t_1, \cdots, t_n\} \subset T$ the finite dimensional distributions are the joint distributions
$$F(x_{t_1}, \cdots, x_{t_n}) = P(X_{t_1} \leq x_{t_1}, \cdots, X_{t_n} \leq x_{t_n})$$

3.1.2 White Noise

- A sequence of uncorrelated random variables $\{X_t\}$, each with $E(X_t) = 0$ and $Var(X_t) = \sigma^2$, for all $t \in T$, is called a white noise process
- Denoted as $WN(0, \sigma^2)$

3.1.3 Gaussian Process

- Let $T = \mathbb{Z}$, then $\{X_t\}$ is a discrete Gaussian process if any finite subset $\{X_{t_1}, \cdots, X_{t_n}\}$ has an n-dimensional multivariate normal distribution
- This model is completely determined when the mean and variance-covariance structures are known

3.1.4 Random Walk

- Let Z_t, $t \in \mathbb{Z}$, be an iid sequence of random variables
- The series defined by
$$X_t = \sum_{i=1}^{t} Z_i$$
for $t = 1, 2, \cdots$ is called a random walk

3.1.5 MA(1) Process

- Let Z_t, $t \in \mathbb{Z}$, be an iid sequence of $N(0, \sigma^2)$, or more generality $Z_t \sim WN(0, \sigma^2)$, random variables
- The series defined by
$$X_t = Z_t + \sigma Z_{t-1}$$
for all t, is called a first order moving average process, and denoted by $MA(1)$

3.1.6 AR(1) Process

- Let Z_t, $t \in \mathbb{Z}$, be an iid sequence of $N(0, \sigma^2)$, or more generality $Z_t \sim WN(0, \sigma^2)$, random variables
- The series defined by
$$X_t = \phi X_{t-1} + Z_t$$
for all $t \in \mathbb{Z}$, is called a first order autoregressive process $AR(1)$

3.2 Stationary Processes

3.2.1 The auto-covariance function

- If $\{X_t\}$ is a time series with $Var(X_t) < \infty$ for all $t \in T$
- The auto-covariance function is defined by
$$\gamma(r, s) = Cov(X_r, X_s)$$
for $r, s \in T$

3.2.2 Stationarity

The time series $\{X_t\}_{t \in T}$ is said to be stationary if

- $E(|X_t|^2) < \infty$ for all $t \in T$
- $E(X_t) = \mu$ for all $t \in T$
- the auto-covariance function satisfies
$$\gamma(r, s) = \gamma(r + t, s + t)$$
for all $r, s, r + t, s + t \in T$

3.2.3 Strict Stationarity

The time series $\{X_t\}_{t \in T}$ is said to be strictly stationary if the finite dimensional vectors $(X_{t_1}, \cdots, X_{t_n})$ and $(X_{t_1+h}, \cdots, X_{t_n+h})$ have the same joint distributions for all finite subsets of T and all h where the translation is defined

3.2.4 Properties of Auto-Covariance Function

Assume $\gamma(r, s)$ is the auto-covariance function of a stationary process $\{X_t\}$, then the following statements hold

- The auto-covariance function can be written as $\gamma(h) = \gamma(r + h, r)$ for all r
- $\gamma(0) \geq 0$
- $|\gamma(h)| \leq \gamma(0)$
- the auto-covariance function is even function, $\gamma(h) = \gamma(-h)$

3.2.5 Auto-Correlation Function

For a stationary process $\{X_t\}$ the auto-correlation function is defined by

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

3.3 MA(q) Processes

3.3.1 Backward Shift Operator

The backward shift operator B acts on a time series $\{X_t\}$ and is defined as

$$BX_t = X_{t-1}$$

The difference operator Δ is also define as $\{X_t\}$ via

$$\Delta X_t = X_t - X_{t-1} = (1 - B)X_t$$

where 1 here represents the identity operator

- We can combine operators as 'polynomials', thus

$$B^0 X_t = X_t, BX_t = X_{t-1}, B^2 X_t = X_{t-2}, \cdots$$

- Define

$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$$

- Thus

$$\theta(B)X_t = X_t + \theta_1 X_{t-1} + \cdots + \theta_q X_{t-q}$$

3.3.2 MA(q) Process

Let $\{Z_t\}$, $t \in \mathbb{Z}$, be an iid sequence of $N(0, \sigma^2)$, or $Z_t \sim WN(0, \sigma^2)$, random variables
The series defined by

$$X_t = \theta(B)Z_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

for all t, is called q-th order moving average, denoted by $MA(q)$

Theorem

Consider a $MA(1)$ process of the form $X_t = Z_t + \theta Z_{t-1}$, for $Z_t \overset{iid}{\sim} N(0, \sigma^2)$, random variables, then we have the following results

- the auto-covariance function is defined by

$$\gamma(r, s) = \begin{cases} \sigma^2(1 + \theta^2) & \text{if } r = s \\ \sigma^2\theta & \text{if } |r - s| = 1 \\ 0 & \text{otherwise} \end{cases}$$

- X_t is a stationary process for all values of θ and $\sigma > 0$

3.3.3 Moments of MA(q) Process

- The mean of an $MA(q)$ process is

$$E(X_t) = E(\theta(B)Z_t) = 0$$

  for all t

- The auto-covariance of an $MA(q)$ process is given by

$$\gamma(r,s) = \begin{cases} \sigma^2 \sum_{j=0}^{q-|r-s|} \theta_j \theta_{j+|r-s|} & \text{if } |r-s| \leq q \\ 0 & \text{otherwise} \end{cases}$$

- All $MA(q)$ processes are stationary

3.4 MA(∞) Process

- Let Z_t, $t \in \mathbb{Z}$, be an iid sequence of $N(0, \sigma^2)$, random variables
- Let $\{\psi_j\}, j = 0, 1, \cdots$ be a sequence which is absolutely convergent

$$\sum_{j=0}^{\infty} |\psi| < \infty$$

- The process defined by

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

  for all t, is called an infinite order moving average, and denoted by $MA(\infty)$

The $MA(\infty)$ process is stationary with zero mean and auto-covariance function

$$\gamma(h) = \sigma^2 \sum_{j=-\infty}^{\infty} \psi_j \psi_{j+h}$$

3.4.1 Wold Decomposition Theorem

- Any stationary process can be written as the sum of an $MA(\infty)$ process and an independent determininistic process
- A deteerministic process is any process whose complete realisation is a deterministic function of a finite number of its value

3.5 AR(p) Process

- Define the polynomial operator

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

  where B is the backward shift operator

- The $AR(p)$ process is the process which is the stationary solution to the difference equations

$$\phi(B)X_t = Z_t$$

  for $\{Z_t\} \sim WN(0, \sigma^2)$, when such a solution exists

3.5.1 Causal Process

- A causal process $\{X_t\}$ generated by

$$\{Z_t\} \sim WN(0, \sigma^2)$$

  is one where each X_t is only a function of those Z_s where $s \leq t$

- It seems intuitive to only consider models where the current value is a function of past values

3.5.2 Existence of Stationary Solution

- There exists a stationary solution to $\phi(B)X_t = Z_t$ when, for $z \in \mathbb{C}$, the polynomial

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$$

  has no roots which which lie on the unit $\{z \in \mathbb{C} | z| = 1\}$

- If all roots lie strictly outside the unit circle we say there is a causal solution which can be written as

$$X_t = \sum_{j=1}^{\infty} \phi_j Z_{t-j}$$

3.5.3 Auto-Covariance Function for AR(p) Process

If X_t is a causal, stationary $AR(p)$ process it can be written as

$$X_t = \sum_{j=0}^{\infty} \phi_j Z_{t-j}$$

and hence its auto-covariance function can be written as

$$\gamma(h) = \sigma^2 \sum_{j=0}^{\infty} \phi_j \phi_{j+|h|}$$

3.5.4 AR(1) Process

- Let $\{X_t\}$ be an AR(1) process defined by

$$X_t = \phi X_{t-1} + Z_t$$

  for $Z_t \overset{iid}{\sim} N(0, \sigma^2)$, random variables

- When the process is stationary an causal then the auto-covariance function is given by

$$\gamma(h) = \frac{\sigma^2 \phi^{|h|}}{1 - \phi^2}$$

3.6 Partial ACVF

The partial auto-covariance function (PACF) for a stationary process $\{X_t\}$ is defined by

$$\alpha(h) = \begin{cases} Cor(X_1, X_0) & \text{for } |h| = 1 \\ \rho x_h x_0 \cdot \{x_{h-1} \cdots x_1\} & \text{for } |h| > 1 \end{cases}$$

where $\rho x_h x_0 \cdot \{x_{h-1} \cdots x_1\}$ is the partial correlation of X_h and X_0 given the set $\{X_{h-1} \cdots X_1\}$

3.6.1 Partial Auto-Correlation Function

If $\alpha(h)$ is the partial auto-covariance function for a stationary $AR(p)$ process then $\alpha(k) = 0$ for $|k| > p$

3.7 ARMA(p,q) Process

Let $Z_t \sim WN(0, \sigma^2)$, random variables, and define the polynomial operators

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$$

and

$$\theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q$$

where B is the backward shift operator.
The process is the stationary solution to the difference equations

$$\phi(B) X_t = \theta(B) Z_t$$

when such a solution exists, and we assume that the polynomials $\phi(z)$ and $\theta(z)$ share no common factors

3.7.1 Existence of ARMA Process

The ARMA(p,q) process defined by $\phi(B)X_t = \theta(B)Z_t$ has a stationary solution if none of the roots of polynomial $\phi(z)$ lie on the unit circle $|z| = 1$. Further, if the roots lie outside the unit circle then there iis a causal stationary solution

4 Model Based Forecasting for Time Series

4.1 Best Linear Predictor

4.1.1 Best Linear Predictor

Consider two random variables X and Y, with $E(X) = \mu_X$ and $E(Y) = \mu_Y$ and all second moments finite
The best linear predictor is

$$\hat{Y} = E(Y) + \frac{Cov(X,Y)}{Var(X)}(X - \mu_X)$$

The MSE of the predictor is

$$Var(Y)(1 - Corr(X,Y)^2)$$

Note that if X and Y are uncorrelated, the best linear predictor of Y is

$$\hat{Y} = E(Y)$$

4.1.2 Non-Linear Prediction

Suppose $X \sim N(0,1)$ and $Y = X^2 - 1$, then we have

$$E(X) = E(Y) = 0$$

and the best linear predictor is $\hat{Y} = 0$, since $Cov(X,Y) = 0$
The best predictor of Y given X is $X^2 - 1$, which has MSE of 0, best linear predictor might not be very good

4.1.3 Best Linear Predictor

Suppose we want to predict Y given X_1, X_2. The best linear predictor is

$$\hat{Y} = \alpha_0 + \alpha_1 X_2 + \alpha_2 X_1$$

When covariance is non-singular the best predictor is

$$E(Y) + \begin{bmatrix} X_2 - \mu X_2 & X_1 - \mu X_1 \end{bmatrix} \begin{bmatrix} Var(X_2) & Cov(X_1, X_2) \\ Cov(X_1, X_2) & Var(X_1) \end{bmatrix}^{-1} \begin{bmatrix} Cov(Y, X_2) \\ Cov(Y, X_1) \end{bmatrix}$$

4.1.4 Prediction Operator

- If $\{X_t\}$ is a stationary time series, with mean μ and auto-covariance function $\gamma(h)$
- The best linear predictor of X_{n+h} given the set $\{X_n, \cdots, X_1\}$ is

$$Pred(X_{n+h}|\{X_n, \cdots, X_1\}) = \mu + \begin{bmatrix} \hat{\alpha_1} & \cdots & \hat{\alpha_n} \end{bmatrix} \begin{bmatrix} X_n - \mu \\ \vdots \\ X_1 - \mu \end{bmatrix}$$

- where $\begin{bmatrix}\hat{\alpha_1} & \cdots & \hat{\alpha_n}\end{bmatrix}$ satisfies the equation

$$\Gamma \begin{bmatrix}\hat{\alpha_1} \\ \vdots \\ \hat{\alpha_n}\end{bmatrix} = \begin{bmatrix}\gamma(h) \\ \vdots \\ \gamma(h+n-1)\end{bmatrix}$$

- Here Γ is the $n \times n$ matrix with ij-th element

$$\Gamma_{ij} = \gamma(|i-j|)$$

- The MSE is given by

$$\gamma(0) - \hat{\alpha}^T \gamma_{n,h}$$

This theorem shows that for linear prediction for a stationary process $\{X_t\}$ only need

- mean of the process

$$E(X_t) = \mu$$

- The auto-correlation function $\gamma(h)$ for a suitable range of h

4.1.5 Theorem

Let U be a random variable, such that $E(U^2) < \infty$, and W an n-dimensional random vector with a finite variance-covariance matrix

$$\Gamma = Cov(W, W)$$

In terms of mean square error the best linear predictor, $Pred(U|W)$, of U based on $W = (W_1, \cdots, W_n)$ satisfies

1. $Pred(U|W) = E(U) + \alpha^T(W - E(W))$ where $\Gamma\alpha = Cov(U, W)$
2. $E([U - Pred(U|W)]) = 0$, the estimator is unbiased
3. $E([U - Pred(U|W)]W) = 0$, the zero vector
4. $Pred(\alpha_0 + \alpha_1 U_1 + \alpha_2 U_2|W) = \alpha_0 + \alpha_1 Pred(U_1|W) + \alpha_2 Pred(U_2|W)$, the predictor is linear
5. $Pred(W_i|W) = W_i$

4.2 Model Based Approach

- Let $x_1, \cdots, x_n$ be observed values of a stationary time series
- The sample mean is defined by

$$\overline{x} = \frac{1}{n}\sum_{t=1}^{n} x_t$$

- The sample auto-covariance function is defined by

$$\hat{(}\gamma)(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(x_{t+|h|} - \overline{x})(x_t - \overline{x})$$

and the sample auto-correlation function is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

4.3 Estimation

4.3.1 The Likelihood Function

The likelihood function for a mean zero, stationary normal, $ARMA(p, q)$ process is

$$L(\mu, \phi, \theta, \sigma^2) = \frac{1}{(2\pi)^{n/2} det(\Gamma_n)^{1/2}} \exp\left(-\frac{1}{2}(X-\mu)^T \Gamma_n^{-1}(X-\mu)\right)$$

Γ_n is variance-covariance matrix for X of the form

$$\begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix}$$

where $\gamma(h)$ is the auto-covariance function of the process

4.3.2 AIC

The Akaike information criterion (AIC) is defined as

$$AIC = -2l(\hat{\beta}) + 2k$$

where $l(\hat{\beta})$ is the maximum value of the log-likelihood for a given model, and k is the number of parameters in the model
The AICc corrects for small samples and is given by

$$AICc = AIC + \frac{2k(k+1)}{n-k-1}$$

For both criteria in a given set of models the preferred model is the one with the minimum criterion value
The idea is to penalise more complicated model and reward simple models, as long as they fit the data

4.4 The Box-Jenkins Approach

Theorem

- Let U be a random variable, such that $E(U^2) < \infty$
- W an n-dimensional random vector with a finite variance-covariance matrix

$$\Gamma = Cov(W, W)$$

- In terms of mean square error the best linear predictor, $Pred(U|W)$, of U based on $W = (X_1, \cdots, X_n)$ satisfies the following results
  1. $Pred(U|W) = E(U) + a^T(W - E(W))$ where $\Gamma a = Cov(U, W)$
  2. $E([U - Pred(U|W)]) = 0$, the estimator is unbiased
  3. $E([U - Pred(U|W)]W) = 0$, the zero vector
  4. $Pred(\alpha_0 + \alpha_1 U_1 + \alpha_2 U_2|W) = \alpha_0 + \alpha_1 Pred(U_1|W) + \alpha_2 Pred(U_2|W)$, the predictor is linear
  5. $Pred(W_i|W) = W_i$

When a time series $\{X_t\}$ is stationary the best linear predictor $Pred(X_{n+h}|X_1, \cdots, X_n)$ is a function of the mean μ and auto-covariance function $\gamma(\cdot)$

4.4.1 D-th Order Differencing

The operator ∇^d is called the d-th order differencing operator and is defined by

$$\nabla^d X_t = (1 - B)^d X_t$$

4.4.2 ARIMA Model

- Let d be a non-negative integer
- $\{X_t\}$ is an $ARIMA(p, d, q)$ process if

$$Y_t = \nabla^d X_t$$

  is a causal $ARMA(p, q)$ process

4.4.3 Seasonal Components

Let $Z_t \sim WN(0, \sigma^2)$ and, for ease of interpretation, let us assume t is recorded on a yearly time scale, but we have one observation per month
Let α_i, $i = 1, \cdots, 12$, be constants associate with the month $(i-1)/12$ such that $\sum_{i=1}^{12} \alpha_i = 0$, and define the function $mon(t)$ as the function which returns the month associated with the time t
The process $X_t = \alpha_{mon(t)} + Z_t$ where $\{Z_t\} \sim ARIMA(p, d, q)$ is said to have a seasonal component. It will not be stationary if any of the α_i values are non-zero

4.4.4 Lag s Difference

The lag s difference operator B^s is defined by

$$B^s X_t = X_{t-s}$$

The corresponding difference operator $\triangle^s$ is defined by

$$\triangle^s = (1 - B^s)$$

4.4.5 Seasonal ARIMA Model

If d and D are non-negative integers, then $\{X_t\}$ is a seasonal $ARIMA(p, d, q)(P, D, Q)$ process with period s if the differenced series

$$Y_t = (1 - B)^d (1 - B^s)^D X_t$$

is a causal $ARMA$ process defined by

$$\phi(B)\Phi(B_s)Y_t = \theta(B)\Theta(B^s)Z_t$$

Here $Z_t \sim WN(0, \sigma^2)$ and

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$$
$$\Phi(z) = 1 - \Phi_1 z - \cdots - \Phi_P z^P$$
$$\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$$
$$\Theta(z) = 1 + \Theta_1 z + \cdots + \Theta_Q z^Q$$

The corresponding process is causal iff all the roots of $\phi(z)$ and $\Phi(z)$ lie outside the unit circle

5 Forecast Evaluation

6 Introduction to Bayesian Forecasting

6.1 Introduction

In Bayesian approach all statements about uncertainty must be well-defined probability statements Forcasting then means computing the conditional probability

$$Pr(Y|D = d, M)$$

given a model M with parameters θ
We do this by computing the probability $Pr(Y, \theta|D = d, M)$ annd the marginalising out the random variable θ

6.2 Bayesian Inference

- We assume that we have a model for the data $f(x|\theta)$
- We want to learn about the vector of parameters θ
- In Bayesian statistics everything is considered a random variable
- All statements about uncertainty are made using the language and calculus or probability theory
- $Pr(\theta)$ which we shall call the prior distribution, what we know about θ before we see the data
- To see what we know about θ after we see the data we apply Bayes theorem to give

$$Pr(\theta|data) = \frac{Pr(data|\theta)Pr(\theta)}{Pr(data)}$$

6.2.1 Predictive Distribution

- The solution to a forecasting problem to forecast X_{new} based on an observed sample $X_1, \cdots, X_n$ which comes from a model $f(x|\theta)$ shoudl be the conditional distribution

$$P(X_{new}|X_1, \cdots, X_n)$$

- This is called the predictive distribution
- It is calculated from the posterior by

$$P(X_{new}|X_1, \cdots, X_n) = \int_\theta P(X_{new}|\theta)P(\theta|X_1, \cdots, X_n)d\theta$$

- In words we average over the model according to the posterior belief in the value of θ given the observed data

6.2.2 Conjugate Priors

A conjugate prior is a prior which, for a given model, have the property that the posterior lies in the same functional family as the prior. The updating using the data only changes the parameter values in the conjugate family

6.2.3 Markov Chain Monte Carlo (MCMC)

- Suppose we have a complex distribution $P(\theta)$ that is known up to a constant
- Rather than integrating directly with the distribution we can compute most statistical functions - such as means, variances, marginal distributions - using a large sample

$$\theta_1, \theta_2, \cdots$$

- To make the sampling fast we don't compute an iid sample
- Rather construct a dependent Markov chain whose stationary (equilibrium) distribution is proportional to $P(\theta)$

6.2.4 Building a Predictive Sample

If we want do prediction using a posterior for θ we use a similar idea. We can take a sampl from the predictive distributino in two steps:

- Draw a sample $\theta^{(i)}$ from the posterior $P(\theta|data)$
- Given $\theta^{(i)}$ draw a prediction from the model $f(y|\theta^{(i)})$ We repeat this, independently fro many different values of i

6.2.5 Bayes and Ridge Regression

Suppose we have the model for Y conditionally on X as

$$Y_i|X, \beta \sim N(\beta_0 + x_i^T\beta, \sigma^2)$$

with independent priors $\beta_j \sim N(0, \tau^2)$. Then, with τ, σ assumed known the log-posterior for β is given

$$\sum_{i=1}^{N}(y_i - \beta_0 + \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

which agrees with $RSS(\lambda)$ when $\lambda = \sigma^2/\tau^2$

7 The Kalman Filter

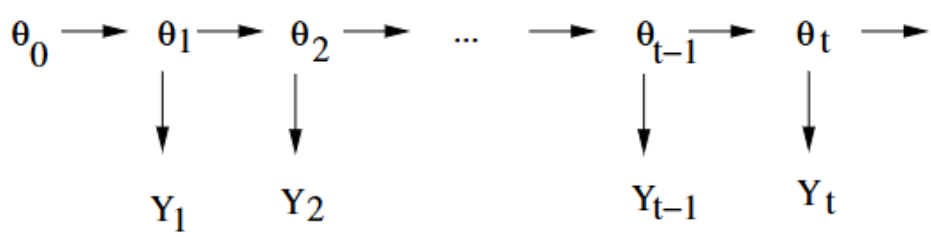
7.1 Introduction

- It is a recursive algorithm which can run in real time - same rate that the data is collected
- it is also an efficient algorithm for computing the likelihood function for arima function
- The Kalman filter allows other sources of information such as known physical laws
- Another important feature is its numerical efficiency
- This allows it to be used in real time control problems such as flying or tracking

7.2 State Space Models

7.2.1 State Space Model

- Suppose $\{Y_t\}$ is an observed time series with unobserved state process $\{\theta_t\}$
- In a state space model the dependence between these is defined byy the graphical shown by



- The relationship defines the conditional independence relations:
  - θ_t is a Markov chain
  - conditionally on $\{\theta_t\}, \{Y_t\}$ are independent
- The joint distribution of $(Y_1, \cdots, Y_t)|(\theta_1, \cdots, \theta_t)$ is then

$$\prod_{i=1}^{t} f(y_t|\theta_t)$$

- State space models are also called hidden Markov models
- In a general state space model we have an unobserved state variable at time s and a set of observations $y_1, \cdots, y_t$
- We subdivide problems by
  - filtering iss the case where $s = t$

- state prediction is the case where $s > t$
- smoothing is the case where $s < t$

Consider a p-dimensional state variable x_t which satisfies the state equation

$$x_t = \Sigma x_{t-1} + \Upsilon u_t + w_t$$

where u_t are a set of control or input variables and $w_t \overset{iid}{\sim} MVN_p(0, Q)$
The observed part of the syste is defined via the observation equation

$$y_t = A_t x_t + \Gamma u_t + v_t$$

where y is q-dimensional and $v_t \overset{iid}{\sim} MVN_q(0, R)$ and the two noise terms, w, v are independent of one another

7.3 Kalman Filter

For the state space model, we have

$$x_t^{t-1} = \Sigma x_{t-1}^{t-2} + \Upsilon u_t$$

This is the prediction part of the method. It has a MSE which satisfies

$$P_t^{t-1} = \Sigma P_{t-1}^{t-2} \Sigma^T + Q$$

While the update is defined by

$$x_t^t = x_t^{t-1} + K_t(y_t - A_t x_{t-1}^t - \Gamma u_t)$$

It has a MSE which satisfies

$$P_t^t = [I - K_t A_t] P_t^{t-1}$$

where the term K_t is called the Kalman gain and is defined by

$$K_t = P_t^{t-1} A_T [A_t P_t^{t-1} A_t^T + R]^{-1}$$

8 ARCH, GARCH, and Hidden Markov Models

8.1 ARCH and GARCH

8.1.1 ARCH Model

An autoregressive conditional heteroscedasticity (ARCH(p)) model is defined hiercrhically: first define

$$X_t = \sigma_t Z_t$$

where $Z_t \sim N(0,1)$ i.i.d., but treat σ as being random such that

$$\sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + \cdots + \alpha_q X_{t-q}^2$$

$\alpha_0 > 0$, $\alpha_i \geq 0$. So the variance is 'time dependent' - a large value of X_t will result in period of high volatility
We can write X_t as a non-linear function of the previous innovations. It will be stationary and causal

8.1.2 Existence of Stationary Solution

For an $ARCH(1)$ model we have, when $|\alpha_1| < 1$ there is a unique causal staitionary solution of the equations

$$X_t = \sigma_t Z_t$$

It has the moment structure:

$$E(X_t) = E(E(X_t|Z_s, s < t)) = 0$$

$$Var(X_t) = \frac{\alpha_0}{1 - \alpha_1}$$

$$Cov(X_{t+h}, X_t) = E(E(X_{t+h} X_t | Z_s, s < t + h)) = 0$$

for $h > 0$

8.1.3 GARCH Model

The $GARCH(p,q)$ model is defined by $X_t = \sigma_t Z_t$ where $Z_t \sim N(0,1)$ i.i.d. and

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i X_{t-i}^2 + \sum_{i=1}^{q} \beta_i \sigma_{t-i}^2$$

The $GARCH(1,1)$ model is commonly used, is stationary if

$$0 < \alpha_1 + \beta_1 < 1$$

Has uncondtional variance

$$\frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$